

perform such tasks more efficiently.

“To take up jobs that are assistive in nature, robots could just be the right choice to relieve humans from basic administrative tasks.”

Srivastava further reflects, “Robots will play a crucial role in tasks that have high error rates. Automation aims to reduce such errors. Humans can teach and guide robots to perform these tasks without committing mistakes.”

The biggest benefit robots bring to organisations is cost optimisation. Even though they bring marginal cost benefits to companies, the shift towards adopting robots will come sooner than expected.

Concerns and the future

Chawla adds, “Robots are smart; they can be useful almost everywhere. They can carry out tasks better than their human counterparts in extreme environmental conditions. However, the biggest challenge that remains unaddressed is that robots can do

Smart Drones And AI To Drive Businesses In India

➤ Companies that wish to legally use drone technology must obtain necessary permits for drone usage from the government

From e-commerce and healthcare to retail and agriculture, drones are ready to conquer the delivery space with AI technology and hi-end efficiency.

India is one of the largest growing markets in the world for unmanned aerial vehicles (UAVs). According to a report published by BIS Research, by 2021, the Indian UAV market will reach US\$ 885.7 million, while the global market will touch US\$ 22.1 billion.

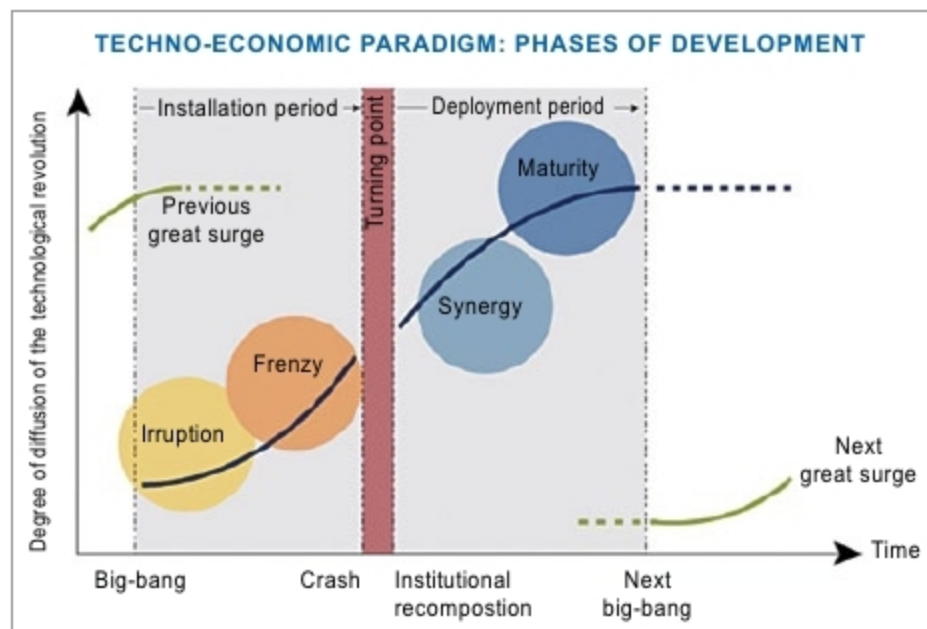
at the pilot phase of testing drone delivery in India. Amazon has filed for a patent for multi-scale fiducials. (Fiducials are black-and-white marks printed on UAVS to help identify these from a distance.) Amazon Prime Air delivery system will allow Amazon to deliver packages within 30 minutes or less in select locations.

Operations for such commercial applications, however, are mostly successful in low-wind and high-visibility areas. Locations with extreme temperatures may not be suitable for usage.

In some remote areas of Africa, drones have proven to be highly successful in providing quick and timely medical aid during a disaster. The operations have saved thousands of lives. Such successful stories are pushing drone adaptation for different industries and geographies.

“Drone technology is the future but, we cannot expect everyone to have access to auto-operated drones, as these are very expensive. We need professionals who understand how to operate drones, their technicalities and features, among others,” says Jagadesh Chander Munivel, director - Maavan Aeronautics, a drone training institute in Chennai.

Indian government has a specific policy for flying drones in the country. It clearly enlists the areas where drones can fly for commercial usage within the geographical boundaries of the country, types of drones allowed along with specified weight categories. Companies that wish to



Adaptation phases for robotics in smart workplaces (Credit: Technological Revolutions and Financial Capital)

repetitive tasks better but, they still cannot handle unexpected circumstances.

“Robots save time but, at the end of the day they are machines. When they malfunction, work could get delayed for much longer than it would with humans.”

Having robots in digital workplaces and making them perform advanced functions is still at an experimental stage. Exact job roles and functions are yet to be defined.

—Nidhi Arora, executive editor, EFY

According to Rahul Papney, lead analyst - BIS Research, India will widely utilise UAVs for both commercial and military applications. However, by 2021, commercial usage will exceed military applications.

In the e-commerce space, giants like Amazon and Flipkart are already



Companies like Amazon and Flipkart are ready to take over the skies through drone delivery (Credit: www.flexport.com)